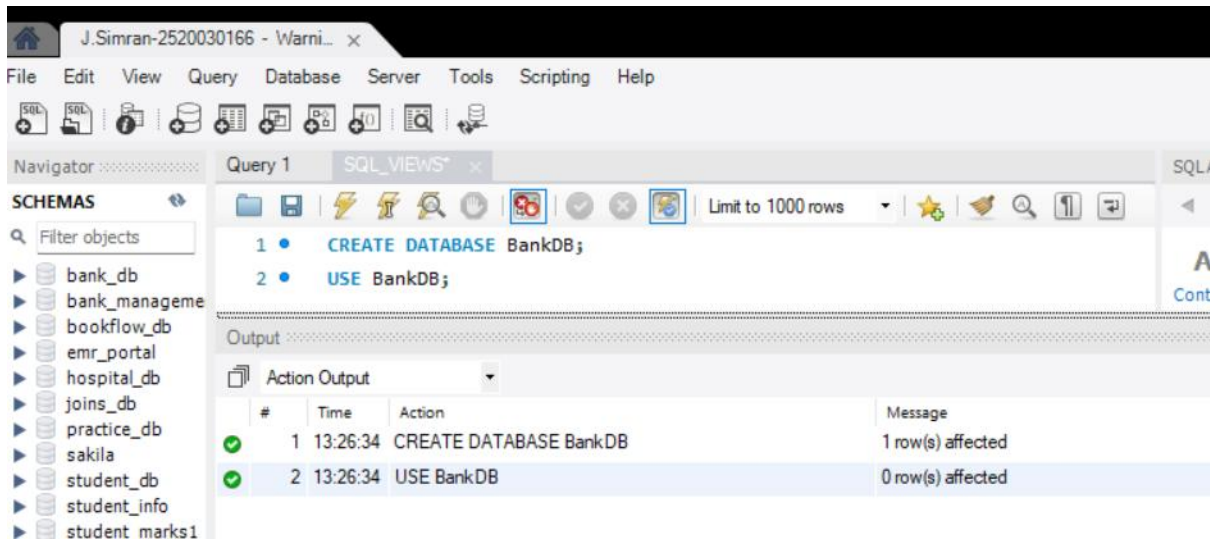


SQL VIEWS

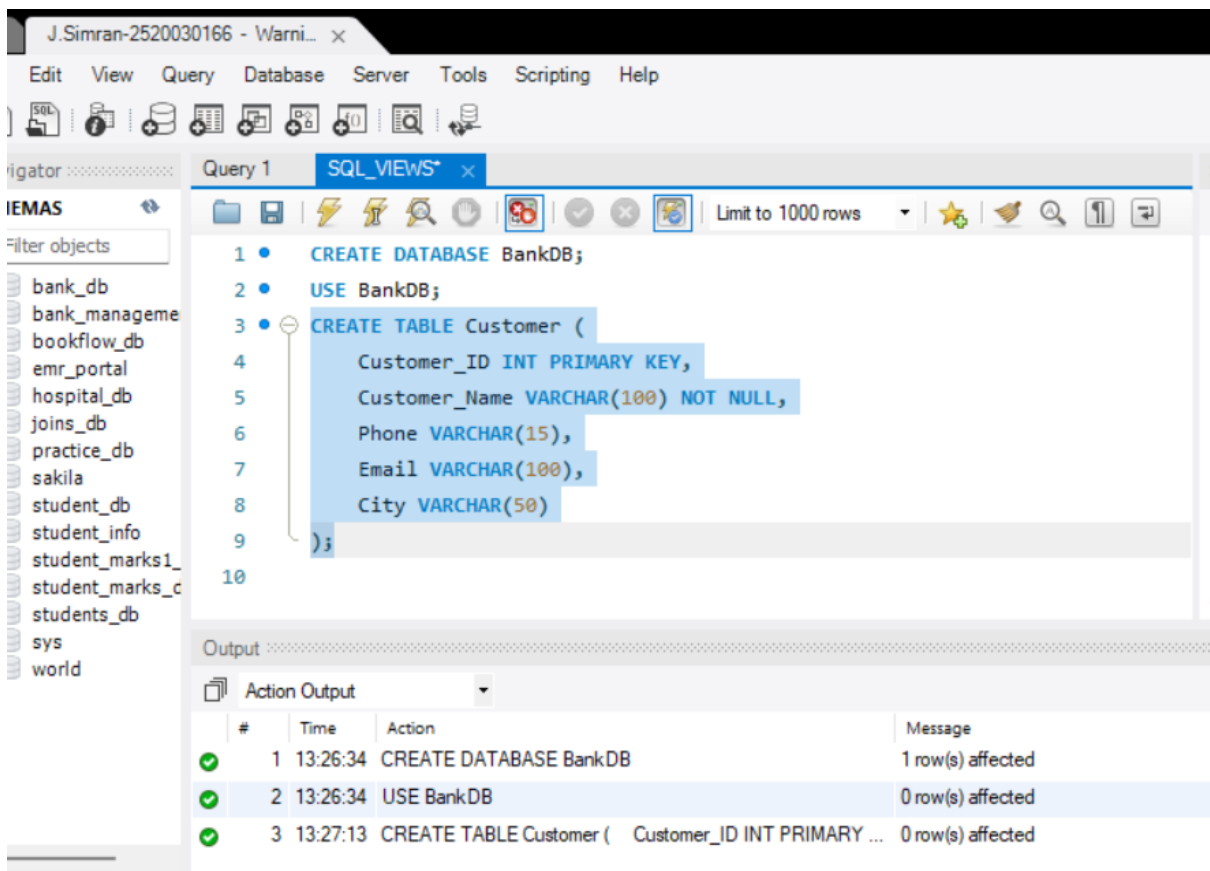
J. Simran

2520030166

1. CREATE DATABASE



2. CREATE CUSTOMER TAB



3. CREATE ACCOUNT TABLE

J.Simran-2520030166 - Warni... x

Edit View Query Database Server Tools Scripting Help

SQL_VIEWS* x

Limit to 1000 rows

```

7      Email VARCHAR(100),
8      City VARCHAR(50)
9  );
10 CREATE TABLE Account (
11     Account_No INT PRIMARY KEY,
12     Customer_ID INT,
13     Account_Type VARCHAR(20),
14     Balance DECIMAL(12,2),
15     Branch VARCHAR(50),
16
17     FOREIGN KEY (Customer_ID)
18     REFERENCES Customer(Customer_ID)
19 );
20

```

Output

Action Output

#	Time	Action	Message
1	13:26:34	CREATE DATABASE BankDB	1 row(s) affected
2	13:26:34	USE BankDB	0 row(s) affected
3	13:27:13	CREATE TABLE Customer (Customer_ID INT PRIMARY ...	0 row(s) affected
4	13:27:53	CREATE TABLE Account (Account_No INT PRIMARY K...	0 row(s) affected

No object selected

4. CREATE BANK TRANSACTION TABLE

J.Simran-2520030166 - Warni... x

Edit View Query Database Server Tools Scripting Help

SQL_VIEWS* x

Limit to 1000 rows

```

17     FOREIGN KEY (Customer_ID)
18     REFERENCES Customer(Customer_ID)
19 );
20 CREATE TABLE Bank_Transaction (
21     Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
22     Account_No INT,
23     Transaction_Type VARCHAR(20),
24     Amount DECIMAL(12,2),
25     Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
26
27     FOREIGN KEY (Account_No)
28     REFERENCES Account(Account_No)
29 );
30

```

Output

Action Output

#	Time	Action	Message
1	13:26:34	CREATE DATABASE BankDB	1 row(s) affected
2	13:26:34	USE BankDB	0 row(s) affected
3	13:27:13	CREATE TABLE Customer (Customer_ID INT PRIMARY ...	0 row(s) affected
4	13:27:53	CREATE TABLE Account (Account_No INT PRIMARY K...	0 row(s) affected
5	13:28:31	CREATE TABLE Bank_Transaction (Transaction_ID INT...	0 row(s) affected

No object selected

5. CREATE LOAN TABLE

The screenshot shows the SQL Developer interface with the SQL editor containing the following SQL script:

```
27 FOREIGN KEY (Account_No)
28 REFERENCES Account(Account_No)
29 );
30 CREATE TABLE Loan (
31 Loan_ID INT PRIMARY KEY,
32 Customer_ID INT,
33 Loan_Type VARCHAR(30),
34 Loan_Amount DECIMAL(12,2),
35 Interest_Rate DECIMAL(5,2),
36
37 FOREIGN KEY (Customer_ID)
38 REFERENCES Customer(Customer_ID)
39 );
40
```

The Output window shows the following actions and messages:

#	Time	Action	Message
1	13:26:34	CREATE DATABASE BankDB	1 row(s) affected
2	13:26:34	USE BankDB	0 row(s) affected
3	13:27:13	CREATE TABLE Customer (Customer_ID INT PRIMARY ...	0 row(s) affected
4	13:27:53	CREATE TABLE Account (Account_No INT PRIMARY K...	0 row(s) affected
5	13:28:31	CREATE TABLE Bank_Transaction (Transaction_ID INT...	0 row(s) affected
6	13:28:58	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected

6. INSERT CUSTOMER DATA

The screenshot shows the SQL Developer interface with the SQL editor containing the following SQL script:

```
50 ('sneha@gmail.com', 'Chennai'),
51 (105, 'Kiran Kumar', '9876543214',
52 'kiran@gmail.com', 'Hyderabad'),
53 (106, 'Anil Kumar', '9876543215',
54 'anil@gmail.com', 'Delhi'),
55 (107, 'Meena Reddy', '9876543216',
56 'meena@gmail.com', 'Mumbai'),
57 (108, 'Rahul Sharma', '9876543217',
58 'rahul@gmail.com', 'Pune'),
59 (109, 'Lakshmi Devi', '9876543218',
60 'lakshmi@gmail.com', 'Hyderabad'),
61 (110, 'Suresh Babu', '9876543219',
62 'suresh@gmail.com', 'Vijayawada');
63
```

The Output window shows the following actions and messages:

#	Time	Action	Message
1	13:26:34	CREATE DATABASE BankDB	1 row(s) affected
2	13:26:34	USE BankDB	0 row(s) affected
3	13:27:13	CREATE TABLE Customer (Customer_ID INT PRIMARY ...	0 row(s) affected
4	13:27:53	CREATE TABLE Account (Account_No INT PRIMARY K...	0 row(s) affected
5	13:28:31	CREATE TABLE Bank_Transaction (Transaction_ID INT...	0 row(s) affected
6	13:28:58	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
7	13:29:47	INSERT INTO Customer (Customer_ID, Customer_Name, Ph...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

7. INSERT ACCOUNT DATA

The screenshot shows the SQL Developer interface with the following components:

- Navigator:** Lists database objects including bank_db, bank_managem, bookflow_db, emr_portal, hospital_db, joins_db, practice_db, sakila, student_db, student_info, student_marks1, student_marks_c, students_db, sys, and world.
- SQL Editor:** Contains the following SQL script:

```
63 INSERT INTO Account
64 (Account_No, Customer_ID, Account_Type, Balance, Branch)
65 VALUES
66 (10001, 101, 'Savings', 50000, 'Hyderabad'),
67 (10002, 102, 'Savings', 75000, 'Vijayawada'),
68 (10003, 103, 'Current', 120000, 'Bangalore'),
69 (10004, 104, 'Savings', 45000, 'Chennai'),
70 (10005, 105, 'Current', 90000, 'Hyderabad'),
71 (10006, 106, 'Savings', 65000, 'Delhi'),
72 (10007, 107, 'Current', 150000, 'Mumbai'),
73 (10008, 108, 'Savings', 35000, 'Pune'),
74 (10009, 109, 'Savings', 85000, 'Hyderabad'),
75 (10010, 110, 'Current', 110000, 'Vijayawada');
```
- Output Window:** Displays the execution results in a table format:

#	Time	Action	Message
1	13:26:34	CREATE DATABASE BankDB	1 row(s) affected
2	13:26:34	USE BankDB	0 row(s) affected
3	13:27:13	CREATE TABLE Customer (Customer_ID INT PRIMARY ...	0 row(s) affected
4	13:27:53	CREATE TABLE Account (Account_No INT PRIMARY K...	0 row(s) affected
5	13:28:31	CREATE TABLE Bank_Transaction (Transaction_ID INT...	0 row(s) affected
6	13:28:58	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
7	13:29:47	INSERT INTO Customer (Customer_ID, Customer_Name, Ph...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
8	13:30:46	INSERT INTO Account (Account_No, Customer_ID, Accou...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0

8. INSERT TRANSACTION DATA

The screenshot shows the SQL Developer interface with the following components:

- Navigator:** Same as the previous screenshot.
- SQL Editor:** Contains the following SQL script:

```
75 (10010, 110, 'Current', 110000, 'Vijayawada');
76 INSERT INTO Bank_Transaction
77 (Account_No, Transaction_Type, Amount)
78 VALUES
79 (10001, 'DEPOSIT', 10000),
80 (10001, 'WITHDRAW', 5000),
81 (10002, 'DEPOSIT', 15000),
82 (10003, 'WITHDRAW', 20000),
83 (10004, 'DEPOSIT', 5000),
84 (10005, 'WITHDRAW', 10000),
85 (10006, 'DEPOSIT', 12000),
86 (10007, 'DEPOSIT', 25000),
87 (10008, 'WITHDRAW', 5000),
88 (10009, 'DEPOSIT', 20000);
```
- Output Window:** Displays the execution results, including the new transaction insert:

#	Time	Action	Message
1	13:26:34	CREATE DATABASE BankDB	1 row(s) affected
2	13:26:34	USE BankDB	0 row(s) affected
3	13:27:13	CREATE TABLE Customer (Customer_ID INT PRIMARY ...	0 row(s) affected
4	13:27:53	CREATE TABLE Account (Account_No INT PRIMARY K...	0 row(s) affected
5	13:28:31	CREATE TABLE Bank_Transaction (Transaction_ID INT...	0 row(s) affected
6	13:28:58	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
7	13:29:47	INSERT INTO Customer (Customer_ID, Customer_Name, Ph...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
8	13:30:46	INSERT INTO Account (Account_No, Customer_ID, Accou...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
9	13:31:49	INSERT INTO Bank_Transaction (Account_No, Transaction...	11 row(s) affected Records: 11 Duplicates: 0 Warnings: 0

9. INSERT LOAN DATA

The screenshot shows the SQL Developer interface with the 'SQL_VIEWS' window open. The SQL editor contains the following code:

```
89 (10010, 'WITHDRAW', 15000);
90 INSERT INTO Loan
91 (Loan_ID, Customer_ID, Loan_Type,
92 Loan_Amount, Interest_Rate)
93 VALUES
94 (501, 101, 'Home Loan', 5000000, 7.5),
95 (502, 102, 'Education Loan', 1000000, 6.5),
96 (503, 103, 'Car Loan', 800000, 8.2),
97 (504, 104, 'Personal Loan', 500000, 10.5),
98 (505, 105, 'Home Loan', 4000000, 7.2),
99 (506, 106, 'Car Loan', 900000, 8.5),
100 (507, 107, 'Business Loan', 3000000, 9.0),
101 (508, 109, 'Personal Loan', 600000, 10.0);
102
```

The 'Output' window shows the execution results:

#	Time	Action	Message
1	13:26:34	CREATE DATABASE BankDB	1 row(s) affected
2	13:26:34	USE BankDB	0 row(s) affected
3	13:27:13	CREATE TABLE Customer (Customer_ID INT PRIMARY ...	0 row(s) affected
4	13:27:53	CREATE TABLE Account (Account_No INT PRIMARY K...	0 row(s) affected
5	13:28:31	CREATE TABLE Bank_Transaction (Transaction_ID INT...	0 row(s) affected
6	13:28:58	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
7	13:29:47	INSERT INTO Customer (Customer_ID, Customer_Name, Ph...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
8	13:30:46	INSERT INTO Account (Account_No, Customer_ID, Accou...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
9	13:31:49	INSERT INTO Bank_Transaction (Account_No, Transactio...	11 row(s) affected Records: 11 Duplicates: 0 Warnings: 0
10	13:33:39	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, L...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0

10. DISPLAY ORIGINAL TABLES

The screenshot shows the SQL Developer interface with the 'SQL_VIEWS' window open. The SQL editor contains the following code:

```
96 (503, 103, 'Car Loan', 800000, 8.2),
97 (504, 104, 'Personal Loan', 500000, 10.5),
98 (505, 105, 'Home Loan', 4000000, 7.2),
99 (506, 106, 'Car Loan', 900000, 8.5),
100 (507, 107, 'Business Loan', 3000000, 9.0),
101 (508, 109, 'Personal Loan', 600000, 10.0);
102 SELECT * FROM Customer;
```

The 'Result Grid' window shows the data from the Customer table:

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad

The 'Output' window shows the execution results:

#	Time	Action	Message
1	13:26:34	CREATE DATABASE BankDB	1 row(s) affected
2	13:26:34	USE BankDB	0 row(s) affected
3	13:27:13	CREATE TABLE Customer (Customer_ID INT PRIMAR...	0 row(s) affected
4	13:27:53	CREATE TABLE Account (Account_No INT PRIMARY ...	0 row(s) affected
5	13:28:31	CREATE TABLE Bank_Transaction (Transaction_ID IN...	0 row(s) affected
6	13:28:58	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
7	13:29:47	INSERT INTO Customer (Customer_ID, Customer_Name, P...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
8	13:30:46	INSERT INTO Account (Account_No, Customer_ID, Acco...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0
9	13:31:49	INSERT INTO Bank_Transaction (Account_No, Transactio...	11 row(s) affected Records: 11 Duplicates: 0 Warnings: 0
10	13:33:39	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type, ...	8 row(s) affected Records: 8 Duplicates: 0 Warnings: 0
11	13:35:00	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned

J.Simran-2520030166 - Warni... x

Edit View Query Database Server Tools Scripting Help

SQL_VIEWS*

Limit to 1000 rows

SQL Add...

Filter objects

bank_db
bank_managem...
bookflow_db
emr_portal
hospital_db
joins_db
practice_db
sakila
student_db
student_info
student_marks1
student_marks_c
students_db
sys
world

198 (505, 105, 'Home Loan', 4000000, 7.2),
199 (506, 106, 'Car Loan', 900000, 8.5),
200 (507, 107, 'Business Loan', 3000000, 9.0),
201 (508, 109, 'Personal Loan', 600000, 10.0);
202 • SELECT * FROM Customer;
203 • SELECT * FROM Account;
204 • SELECT * FROM Bank_Transaction;

Result Grid

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad

Account 2 x

Apply Revert Context

Output

Action Output

#	Time	Action	Message
2	13:26:34	USE BankDB	0 row(s) affected
3	13:27:13	CREATE TABLE Customer (Customer_ID INT PRIMA...	0 row(s) affected
4	13:27:53	CREATE TABLE Account (Account_No INT PRIMAR...	0 row(s) affected
5	13:28:31	CREATE TABLE Bank_Transaction (Transaction_ID I...	0 row(s) affected
6	13:28:58	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
7	13:29:47	INSERT INTO Customer (Customer_ID, Customer_Name, ...	10 row(s) affected Records: 10 Duplicates: 0 Wan
8	13:30:46	INSERT INTO Account (Account_No, Customer_ID, Acc...	10 row(s) affected Records: 10 Duplicates: 0 Wan
9	13:31:49	INSERT INTO Bank_Transaction (Account_No, Transact...	11 row(s) affected Records: 11 Duplicates: 0 Wan
10	13:33:39	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type...	8 row(s) affected Records: 8 Duplicates: 0 Wamin
11	13:35:00	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned
12	13:35:35	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned

ject Info Session

J.Simran-2520030166 - Warni... x

Edit View Query Database Server Tools Scripting Help

SQL_VIEWS*

Limit to 1000 rows

SQL Add...

Filter objects

bank_db
bank_managem...
bookflow_db
emr_portal
hospital_db
joins_db
practice_db
sakila
student_db
student_info
student_marks1
student_marks_c
students_db
sys
world

199 (506, 106, 'Car Loan', 900000, 8.5),
200 (507, 107, 'Business Loan', 3000000, 9.0),
201 (508, 109, 'Personal Loan', 600000, 10.0);
202 • SELECT * FROM Customer;
203 • SELECT * FROM Account;
204 • SELECT * FROM Bank_Transaction;
205 • SELECT * FROM Loan;

Result Grid

Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-25 13:31:49
2	10001	WITHDRAW	5000.00	2026-08-25 13:31:49
3	10002	DEPOSIT	15000.00	2026-08-25 13:31:49
4	10003	WITHDRAW	20000.00	2026-08-25 13:31:49
5	10004	DEPOSIT	5000.00	2026-08-25 13:31:49

Bank Transaction 3 x

Apply Revert

Output

Action Output

#	Time	Action	Message
3	13:27:13	CREATE TABLE Customer (Customer_ID INT PRIMA...	0 row(s) affected
4	13:27:53	CREATE TABLE Account (Account_No INT PRIMAR...	0 row(s) affected
5	13:28:31	CREATE TABLE Bank_Transaction (Transaction_ID I...	0 row(s) affected
6	13:28:58	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
7	13:29:47	INSERT INTO Customer (Customer_ID, Customer_Name, ...	10 row(s) affected Records: 10 Du
8	13:30:46	INSERT INTO Account (Account_No, Customer_ID, Acc...	10 row(s) affected Records: 10 Du
9	13:31:49	INSERT INTO Bank_Transaction (Account_No, Transact...	11 row(s) affected Records: 11 Du
10	13:33:39	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type...	8 row(s) affected Records: 8 Duplic
11	13:35:00	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned
12	13:35:35	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
13	13:36:19	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned

ject Info Session

J.Simran-2520030166 - Warni... x

Edit View Query Database Server Tools Scripting Help

SQL_VIEWS*

Limit to 1000 rows

```

100 (507, 107, 'Business Loan', 3000000, 9.0),
101 (508, 109, 'Personal Loan', 600000, 10.0);
102 SELECT * FROM Customer;
103 SELECT * FROM Account;
104 SELECT * FROM Bank_Transaction;
105 SELECT * FROM Loan;
106

```

Result Grid

Loan_ID	Customer_ID	Loan_Type	Loan_Amount	Interest_Rate
501	101	Home Loan	5000000.00	7.50
502	102	Education Loan	1000000.00	6.50
503	103	Car Loan	800000.00	8.20
504	104	Personal Loan	500000.00	10.50
505	105	Home Loan	4000000.00	7.20

Loan 4 x

Output

Action Output

#	Time	Action	Message
4	13:27:53	CREATE TABLE Account (Account_No INT PRIMAR...	0 row(s) affected
5	13:28:31	CREATE TABLE Bank_Transaction (Transaction_ID I...	0 row(s) affected
6	13:28:58	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
7	13:29:47	INSERT INTO Customer (Customer_ID, Customer_Name, ...	10 row(s) affected Records: 10 Duplicate
8	13:30:46	INSERT INTO Account (Account_No, Customer_ID, Acc...	10 row(s) affected Records: 10 Duplicate
9	13:31:49	INSERT INTO Bank_Transaction (Account_No, Transact...	11 row(s) affected Records: 11 Duplicate
10	13:33:39	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type....	8 row(s) affected Records: 8 Duplicate
11	13:35:00	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned
12	13:35:35	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
13	13:36:19	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned
14	13:36:35	SELECT * FROM Loan LIMIT 0, 1000	8 row(s) returned

PART A – SIMPLE VIEWS

11. CREATE A VIEW FOR ALL CUSTOMERS

J.Simran-2520030166 - Warni... x

Edit View Query Database Server Tools Scripting Help

SQL_VIEWS*

Limit to 1000 rows

```

104 SELECT * FROM Bank_Transaction;
105 SELECT * FROM Loan;
106 CREATE VIEW Customer_View AS
107 SELECT *
108 FROM Customer;
109 SELECT * FROM Customer_View;
110

```

Result Grid

Customer_ID	Customer_Name	Phone	Email	City
101	Ravi Kumar	9876543210	ravi@gmail.com	Hyderabad
102	Priya Sharma	9876543211	priya@gmail.com	Vijayawada
103	Arjun Reddy	9876543212	arjun@gmail.com	Bangalore
104	Sneha Rao	9876543213	sneha@gmail.com	Chennai
105	Kiran Kumar	9876543214	kiran@gmail.com	Hyderabad
106	Anil Kumar	9876543215	anil@gmail.com	Delhi
107	Meena Reddy	9876543216	meena@gmail.com	Mumbai
108	Rahul Sharma	9876543217	rahul@gmail.com	Pune
109	Lakshmi Devi	9876543218	lakshmi@gmail.com	Hyderabad

Customer_View 5 x

Output

Action Output

#	Time	Action	Message
6	13:28:58	CREATE TABLE Loan (Loan_ID INT PRIMARY KEY, ...	0 row(s) affected
7	13:29:47	INSERT INTO Customer (Customer_ID, Customer_Name, ...	10 row(s) affected Records: 10 Duplicate
8	13:30:46	INSERT INTO Account (Account_No, Customer_ID, Acc...	10 row(s) affected Records: 10 Duplicate
9	13:31:49	INSERT INTO Bank_Transaction (Account_No, Transact...	11 row(s) affected Records: 11 Duplicate
10	13:33:39	INSERT INTO Loan (Loan_ID, Customer_ID, Loan_Type....	8 row(s) affected Records: 8 Duplicate
11	13:35:00	SELECT * FROM Customer LIMIT 0, 1000	10 row(s) returned
12	13:35:35	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
13	13:36:19	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned

12. CREATE A VIEW FOR CUSTOMER BASIC INFORMATION

The screenshot shows the SQL_VIEWS* window in a database management tool. The SQL editor contains the following code:

```
110 • CREATE VIEW Customer_Basic_View AS
111 SELECT
112     Customer_ID,
113     Customer_Name,
114     City
115 FROM Customer;
116 • SELECT * FROM Customer_Basic_View;
```

The Result Grid displays the data from the Customer table:

Customer_ID	Customer_Name	City
101	Ravi Kumar	Hyderabad
102	Priya Sharma	Vijayawada
103	Arjun Reddy	Bangalore
104	Sneha Rao	Chennai
105	Kiran Kumar	Hyderabad
106	Anil Kumar	Delhi
107	Meena Reddy	Mumbai
108	Rahul Sharma	Pune
109	Lakshmi Devi	Hyderabad

The Output window shows the execution results:

#	Time	Action	Message
12	13:35:35	SELECT * FROM Account LIMIT 0, 1000	10 row(s) returned
13	13:36:19	SELECT * FROM Bank_Transaction LIMIT 0, 1000	11 row(s) returned
14	13:36:35	SELECT * FROM Loan LIMIT 0, 1000	8 row(s) returned
15	13:37:55	CREATE VIEW Customer_View AS SELECT * FROM Cus...	0 row(s) affected
16	13:37:55	SELECT * FROM Customer_View LIMIT 0, 1000	10 row(s) returned
17	13:39:27	CREATE VIEW Customer_Basic_View AS SELECT Cu...	0 row(s) affected
18	13:39:27	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	10 row(s) returned

13. CREATE A VIEW FOR ACCOUNT INFORMATION

The screenshot shows the SQL_VIEWS* window in a database management tool. The SQL editor contains the following code:

```
119     Account_No,
120     Account_Type,
121     Balance,
122     Branch
123 FROM Account;
124 • SELECT * FROM Account_View;
125
```

The Result Grid displays the data from the Account table:

Account_No	Account_Type	Balance	Branch
10001	Savings	50000.00	Hyderabad
10002	Savings	75000.00	Vijayawada
10003	Current	120000.00	Bangalore
10004	Savings	45000.00	Chennai
10005	Current	90000.00	Hyderabad
10006	Savings	65000.00	Delhi
10007	Current	150000.00	Mumbai
10008	Savings	35000.00	Pune
10009	Savings	85000.00	Hyderabad

The Output window shows the execution results:

#	Time	Action	Message
14	13:36:35	SELECT * FROM Loan LIMIT 0, 1000	8 row(s) returned
15	13:37:55	CREATE VIEW Customer_View AS SELECT * FROM Cus...	0 row(s) affected
16	13:37:55	SELECT * FROM Customer_View LIMIT 0, 1000	10 row(s) returned
17	13:39:27	CREATE VIEW Customer_Basic_View AS SELECT Cu...	0 row(s) affected
18	13:39:27	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	10 row(s) returned
19	13:40:25	CREATE VIEW Account_View AS SELECT Account_...	0 row(s) affected
20	13:40:25	SELECT * FROM Account_View LIMIT 0, 1000	10 row(s) returned

14. CREATE A VIEW FOR SAVINGS ACCOUNTS

The screenshot shows the SQL Developer interface with the following components:

- SQL_VIEWS* window:** Contains the following SQL script:

```
123 FROM Account;  
124 SELECT * FROM Account_View;  
125 CREATE VIEW Savings_Account_View AS  
126 SELECT *  
127 FROM Account  
128 WHERE Account_Type = 'Savings';  
129 SELECT * FROM Savings_Account_View;
```
- Result Grid:** Displays the data from the Savings_Account_View. The data is as follows:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10004	104	Savings	45000.00	Chennai
10006	106	Savings	65000.00	Delhi
10008	108	Savings	35000.00	Pune
10009	109	Savings	85000.00	Hyderabad
- Output window:** Shows the execution log with the following entries:

#	Time	Action	Message
16	13:37:55	SELECT * FROM Customer_View LIMIT 0, 1000	10 row(s) returned
17	13:39:27	CREATE VIEW Customer_Basic_View AS SELECT	0 row(s) affected
18	13:39:27	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	10 row(s) returned
19	13:40:25	CREATE VIEW Account_View AS SELECT	0 row(s) affected
20	13:40:25	SELECT * FROM Account_View LIMIT 0, 1000	10 row(s) returned
21	13:41:28	CREATE VIEW Savings_Account_View AS SELECT * F...	0 row(s) affected
22	13:41:28	SELECT * FROM Savings_Account_View LIMIT 0, 1000	6 row(s) returned

15. CREATE A VIEW FOR CURRENT ACCOUNTS

The screenshot shows the SQL Developer interface with the following components:

- SQL_VIEWS* window:** Contains the following SQL script:

```
125 CREATE VIEW Savings_Account_View AS  
126 SELECT *  
127 FROM Account  
128 WHERE Account_Type = 'Savings';  
129 SELECT * FROM Savings_Account_View;  
130 CREATE VIEW Current_Account_View AS  
131 SELECT *  
132 FROM Account  
133 WHERE Account_Type = 'Current';  
134 SELECT * FROM Current_Account_View;
```
- Result Grid:** Displays the data from the Current_Account_View. The data is as follows:

Account_No	Customer_ID	Account_Type	Balance	Branch
10003	103	Current	120000.00	Bangalore
10005	105	Current	90000.00	Hyderabad
10007	107	Current	150000.00	Mumbai
10010	110	Current	110000.00	Vijayawada
- Output window:** Shows the execution log with the following entries:

#	Time	Action	Message
18	13:39:27	SELECT * FROM Customer_Basic_View LIMIT 0, 1000	10 row(s) returned
19	13:40:25	CREATE VIEW Account_View AS SELECT	0 row(s) affected
20	13:40:25	SELECT * FROM Account_View LIMIT 0, 1000	10 row(s) returned
21	13:41:28	CREATE VIEW Savings_Account_View AS SELECT * F...	0 row(s) affected
22	13:41:28	SELECT * FROM Savings_Account_View LIMIT 0, 1000	6 row(s) returned
23	13:42:54	CREATE VIEW Current_Account_View AS SELECT * FR...	0 row(s) affected
24	13:42:54	SELECT * FROM Current_Account_View LIMIT 0, 1000	4 row(s) returned

PART B – VIEWS WITH CONDITIONS

16. HIGH BALANCE ACCOUNT VIEW

The screenshot shows the SQL Developer interface with the 'SQL_VIEWS' window open. The left sidebar displays a list of database objects, including 'bank_db', 'bank_managemen', 'bookflow_db', 'emr_portal', 'hospital_db', 'joins_db', 'practice_db', 'sakila', 'student_db', 'student_info', 'student_marks1', 'student_marks_c', 'students_db', 'sys', and 'world'. The main window displays the following SQL script:

```
134 • SELECT * FROM Current_Account_View;
135 • CREATE VIEW High_Balance_View AS
136 SELECT
137     Account_No,
138     Customer_ID,
139     Account_Type,
140     Balance
141 FROM Account
142 WHERE Balance > 100000;
143 • SELECT * FROM High_Balance_View;
```

The 'Result Grid' shows the following data:

Account_No	Customer_ID	Account_Type	Balance
10003	103	Current	120000.00
10007	107	Current	150000.00
10010	110	Current	110000.00

The 'Output' window shows the following messages:

#	Time	Action	Message
20	13:40:25	SELECT * FROM Account_View LIMIT 0, 1000	10 row(s) returned
21	13:41:28	CREATE VIEW Savings_Account_View AS SELECT * F...	0 row(s) affected
22	13:41:28	SELECT * FROM Savings_Account_View LIMIT 0, 1000	6 row(s) returned
23	13:42:54	CREATE VIEW Current_Account_View AS SELECT * FR...	0 row(s) affected
24	13:42:54	SELECT * FROM Current_Account_View LIMIT 0, 1000	4 row(s) returned
25	13:44:03	CREATE VIEW High_Balance_View AS SELECT Acc...	0 row(s) affected
26	13:44:03	SELECT * FROM High_Balance_View LIMIT 0, 1000	3 row(s) returned

17. HYDERABAD ACCOUNT VIEW

The screenshot shows the SQL Developer interface with the 'SQL_VIEWS' window open. The left sidebar displays a list of database objects, including 'bank_db', 'bank_managemen', 'bookflow_db', 'emr_portal', 'hospital_db', 'joins_db', 'practice_db', 'sakila', 'student_db', 'student_info', 'student_marks1', 'student_marks_c', 'students_db', 'sys', and 'world'. The main window displays the following SQL script:

```
139 Account_Type,
140 Balance
141 FROM Account
142 WHERE Balance > 100000;
143 • SELECT * FROM High_Balance_View;
144 • CREATE VIEW Hyderabad_Account_View AS
145 SELECT *
146 FROM Account
147 WHERE Branch = 'Hyderabad';
148 • SELECT * FROM Hyderabad_Account_View;
```

The 'Result Grid' shows the following data:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10005	105	Current	90000.00	Hyderabad
10009	109	Savings	85000.00	Hyderabad

The 'Output' window shows the following messages:

#	Time	Action	Message
22	13:41:28	SELECT * FROM Savings_Account_View LIMIT 0, 1000	6 row(s) returned
23	13:42:54	CREATE VIEW Current_Account_View AS SELECT * FR...	0 row(s) affected
24	13:42:54	SELECT * FROM Current_Account_View LIMIT 0, 1000	4 row(s) returned
25	13:44:03	CREATE VIEW High_Balance_View AS SELECT Acc...	0 row(s) affected
26	13:44:03	SELECT * FROM High_Balance_View LIMIT 0, 1000	3 row(s) returned
27	13:44:46	CREATE VIEW Hyderabad_Account_View AS SELECT *...	0 row(s) affected
28	13:44:46	SELECT * FROM Hyderabad_Account_View LIMIT 0, 1000	3 row(s) returned

18. LOW BALANCE ACCOUNT VIEW

The screenshot shows the SQL Developer interface with a query window titled 'SQL_VIEWS*'. The query is as follows:

```
147 WHERE Branch = 'Hyderabad';
148 • SELECT * FROM Hyderabad_Account_View;
149 • CREATE VIEW Low_Balance_View AS
150 SELECT
151     Account_No,
152     Customer_ID,
153     Balance
154 FROM Account
155 WHERE Balance < 50000;
156 • SELECT * FROM Low_Balance_View;
```

The 'Result Grid' shows the following data:

Account_No	Customer_ID	Balance
10004	104	45000.00
10008	108	35000.00

The 'Output' pane shows the execution log with the following messages:

#	Time	Action	Message
24	13:42:54	SELECT * FROM Current_Account_View LIMIT 0, 1000	4 row(s) returned
25	13:44:03	CREATE VIEW High_Balance_View AS SELECT Acc...	0 row(s) affected
26	13:44:03	SELECT * FROM High_Balance_View LIMIT 0, 1000	3 row(s) returned
27	13:44:46	CREATE VIEW Hyderabad_Account_View AS SELECT *...	0 row(s) affected
28	13:44:46	SELECT * FROM Hyderabad_Account_View LIMIT 0, 1000	3 row(s) returned
29	13:45:35	CREATE VIEW Low_Balance_View AS SELECT Acco...	0 row(s) affected
30	13:45:35	SELECT * FROM Low_Balance_View LIMIT 0, 1000	2 row(s) returned

PART C – VIEWS USING JOINS

19. CUSTOMER + ACCOUNT VIEW

The screenshot shows the SQL Developer interface with a query window titled 'SQL_VIEWS*'. The query is as follows:

```
160 C.Customer_Name,
161 C.City,
162 A.Account_No,
163 A.Account_Type,
164 A.Balance,
165 A.Branch
166 FROM Customer C
167 JOIN Account A
168 ON C.Customer_ID = A.Customer_ID;
169 • SELECT * FROM Customer_Account_View;
```

The 'Result Grid' shows the following data:

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance
101	Ravi Kumar	Hyderabad	10001	Savings	50000.00
102	Priya Sharma	Vijayawada	10002	Savings	75000.00
103	Arjun Reddy	Bangalore	10003	Current	120000.00
104	Sneha Rao	Chennai	10004	Savings	45000.00
105	Kiran Kumar	Hyderabad	10005	Current	90000.00

The 'Output' pane shows the execution log with the following messages:

#	Time	Action	Message
26	13:44:03	SELECT * FROM High_Balance_View LIMIT 0, 1000	3 row(s) returned
27	13:44:46	CREATE VIEW Hyderabad_Account_View AS SELECT *...	0 row(s) affected
28	13:44:46	SELECT * FROM Hyderabad_Account_View LIMIT 0, 1000	3 row(s) returned
29	13:45:35	CREATE VIEW Low_Balance_View AS SELECT Acco...	0 row(s) affected
30	13:45:35	SELECT * FROM Low_Balance_View LIMIT 0, 1000	2 row(s) returned
31	13:46:36	CREATE VIEW Customer_Account_View AS SELECT ...	0 row(s) affected
32	13:46:36	SELECT * FROM Customer_Account_View LIMIT 0, 1000	10 row(s) returned

20. HYDERABAD CUSTOMER ACCOUNT VIEW

The screenshot shows the SQL Developer interface with the following components:

- SQL_VIEWS* window:** Contains the SQL code for creating the view:

```
172 C.Customer_Name,  
173 C.City,  
174 A.Account_No,  
175 A.Account_Type,  
176 A.Balance  
177 FROM Customer C  
178 JOIN Account A  
179 ON C.Customer_ID = A.Customer_ID  
180 WHERE A.Branch = 'Hyderabad';  
181 SELECT * FROM Hyderabad_Customer_Accounts;
```
- Result Grid:** Displays the data for the Hyderabad_Customer_Accounts view:

Customer_Name	City	Account_No	Account_Type	Balance
Ravi Kumar	Hyderabad	10001	Savings	50000.00
Kiran Kumar	Hyderabad	10005	Current	90000.00
Lakshmi Devi	Hyderabad	10009	Savings	85000.00
- Output window:** Shows the execution log with the following entries:

#	Time	Action	Message
28	13:44:46	SELECT * FROM Hyderabad_Account_View LIMIT 0, 1000	3 row(s) returned
29	13:45:35	CREATE VIEW Low_Balance_View AS SELECT Acco...	0 row(s) affected
30	13:45:35	SELECT * FROM Low_Balance_View LIMIT 0, 1000	2 row(s) returned
31	13:46:36	CREATE VIEW Customer_Account_View AS SELECT ...	0 row(s) affected
32	13:46:36	SELECT * FROM Customer_Account_View LIMIT 0, 1000	10 row(s) returned
33	13:47:53	CREATE VIEW Hyderabad_Customer_Accounts AS SEL...	0 row(s) affected
34	13:47:53	SELECT * FROM Hyderabad_Customer_Accounts LIMIT ...	3 row(s) returned

21. CUSTOMER LOAN VIEW

The screenshot shows the SQL Developer interface with the following components:

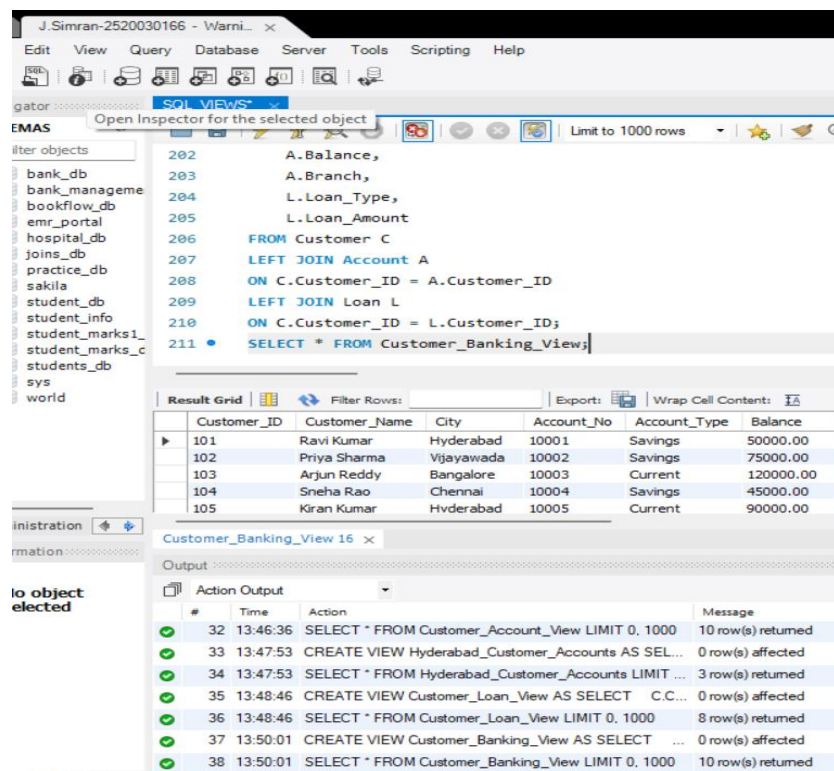
- SQL_VIEWS* window:** Contains the SQL code for creating the view:

```
185 C.Customer_Name,  
186 C.City,  
187 L.Loan_ID,  
188 L.Loan_Type,  
189 L.Loan_Amount,  
190 L.Interest_Rate  
191 FROM Customer C  
192 JOIN Loan L  
193 ON C.Customer_ID = L.Customer_ID;  
194 SELECT * FROM Customer_Loan_View;
```
- Result Grid:** Displays the data for the Customer_Loan_View:

Customer_ID	Customer_Name	City	Loan_ID	Loan_Type	Loan_Amount
101	Ravi Kumar	Hyderabad	501	Home Loan	500000.00
102	Priya Sharma	Vijayawada	502	Education Loan	1000000.00
103	Arjun Reddy	Bangalore	503	Car Loan	800000.00
104	Sneha Rao	Chennai	504	Personal Loan	500000.00
105	Kiran Kumar	Hyderabad	505	Home Loan	400000.00
- Output window:** Shows the execution log with the following entries:

#	Time	Action	Message
30	13:45:35	SELECT * FROM Low_Balance_View LIMIT 0, 1000	2 row(s) returned
31	13:46:36	CREATE VIEW Customer_Account_View AS SELECT ...	0 row(s) affected
32	13:46:36	SELECT * FROM Customer_Account_View LIMIT 0, 1000	10 row(s) returned
33	13:47:53	CREATE VIEW Hyderabad_Customer_Accounts AS SEL...	0 row(s) affected
34	13:47:53	SELECT * FROM Hyderabad_Customer_Accounts LIMIT ...	3 row(s) returned
35	13:48:46	CREATE VIEW Customer_Loan_View AS SELECT C.C...	0 row(s) affected
36	13:48:46	SELECT * FROM Customer_Loan_View LIMIT 0, 1000	8 row(s) returned

22. COMPLETE CUSTOMER BANKING VIEW

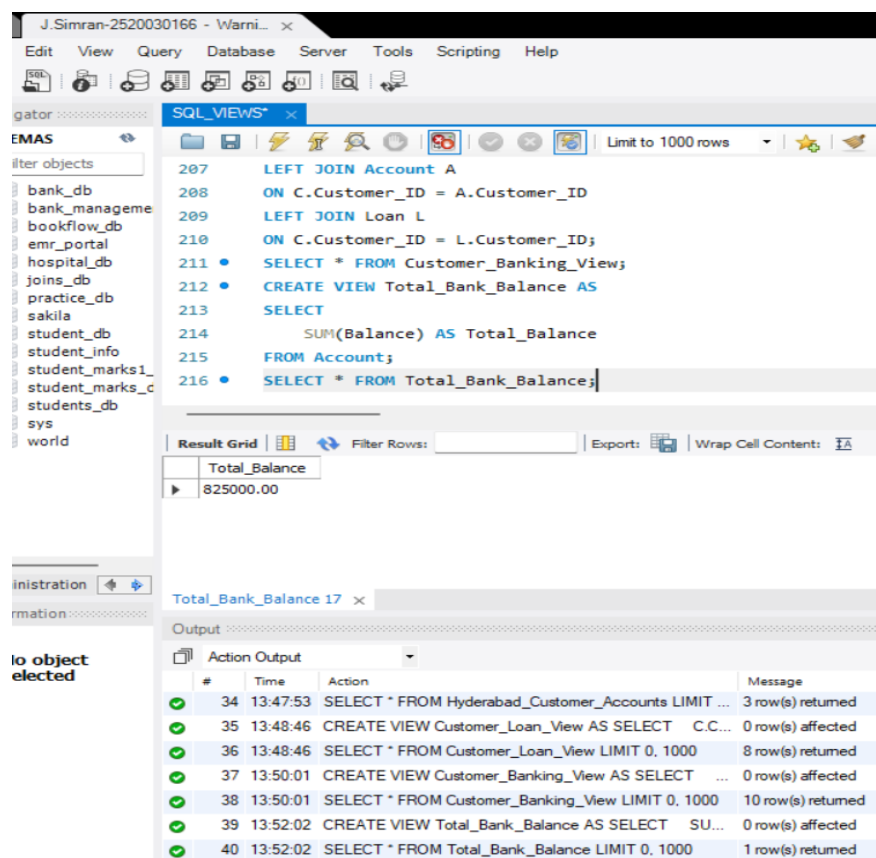


```
202 A.Balance,
203 A.Branch,
204 L.Loan_Type,
205 L.Loan_Amount
206 FROM Customer C
207 LEFT JOIN Account A
208 ON C.Customer_ID = A.Customer_ID
209 LEFT JOIN Loan L
210 ON C.Customer_ID = L.Customer_ID;
211 SELECT * FROM Customer_Banking_View;
```

Customer_ID	Customer_Name	City	Account_No	Account_Type	Balance
101	Ravi Kumar	Hyderabad	10001	Savings	50000.00
102	Priya Sharma	Vijayawada	10002	Savings	75000.00
103	Arjun Reddy	Bangalore	10003	Current	120000.00
104	Sneha Rao	Chennai	10004	Savings	45000.00
105	Kiran Kumar	Hvderabad	10005	Current	90000.00

PART D – VIEWS USING AGGREGATE FUNCTIONS

23. TOTAL BANK BALANCE



```
207 LEFT JOIN Account A
208 ON C.Customer_ID = A.Customer_ID
209 LEFT JOIN Loan L
210 ON C.Customer_ID = L.Customer_ID;
211 SELECT * FROM Customer_Banking_View;
212 CREATE VIEW Total_Bank_Balance AS
213 SELECT
214     SUM(Balance) AS Total_Balance
215 FROM Account;
216 SELECT * FROM Total_Bank_Balance;
```

Total_Balance
825000.00

24. AVERAGE ACCOUNT BALANCE

The screenshot shows the SQL Developer interface with the following SQL script in the SQL_VIEWS* window:

```
212 • CREATE VIEW Total_Bank_Balance AS
213 SELECT
214     SUM(Balance) AS Total_Balance
215 FROM Account;
216 • SELECT * FROM Total_Bank_Balance;
217 • CREATE VIEW Average_Account_Balance AS
218 SELECT
219     AVG(Balance) AS Average_Balance
220 FROM Account;
221 • SELECT * FROM Average_Account_Balance;
```

The Result Grid shows the output of the last query:

Average_Balance
82500.000000

The Action Output window shows the execution log:

#	Time	Action	Message
✓ 36	13:48:46	SELECT * FROM Customer_Loan_View LIMIT 0, 1000	8 row(s) returned
✓ 37	13:50:01	CREATE VIEW Customer_Banking_View AS SELECT ...	0 row(s) affected
✓ 38	13:50:01	SELECT * FROM Customer_Banking_View LIMIT 0, 1000	10 row(s) returned
✓ 39	13:52:02	CREATE VIEW Total_Bank_Balance AS SELECT SU...	0 row(s) affected
✓ 40	13:52:02	SELECT * FROM Total_Bank_Balance LIMIT 0, 1000	1 row(s) returned
✓ 41	13:53:16	CREATE VIEW Average_Account_Balance AS SELECT ...	0 row(s) affected
✓ 42	13:53:16	SELECT * FROM Average_Account_Balance LIMIT 0, 1...	1 row(s) returned

25. MAXIMUM ACCOUNT BALANCE

The screenshot shows the SQL Developer interface with the following SQL script in the SQL_VIEWS* window:

```
217 • CREATE VIEW Average_Account_Balance AS
218 SELECT
219     AVG(Balance) AS Average_Balance
220 FROM Account;
221 • SELECT * FROM Average_Account_Balance;
222 • CREATE VIEW Maximum_Balance_View AS
223 SELECT
224     MAX(Balance) AS Maximum_Balance
225 FROM Account;
226 • SELECT * FROM Maximum_Balance_View;
```

The Result Grid shows the output of the last query:

Maximum_Balance
150000.00

The Action Output window shows the execution log:

#	Time	Action	Message
✓ 38	13:50:01	SELECT * FROM Customer_Banking_View LIMIT 0, 1000	10 row(s) returned
✓ 39	13:52:02	CREATE VIEW Total_Bank_Balance AS SELECT SU...	0 row(s) affected
✓ 40	13:52:02	SELECT * FROM Total_Bank_Balance LIMIT 0, 1000	1 row(s) returned
✓ 41	13:53:16	CREATE VIEW Average_Account_Balance AS SELECT ...	0 row(s) affected
✓ 42	13:53:16	SELECT * FROM Average_Account_Balance LIMIT 0, 1...	1 row(s) returned
✓ 43	13:54:35	CREATE VIEW Maximum_Balance_View AS SELECT ...	0 row(s) affected
✓ 44	13:54:35	SELECT * FROM Maximum_Balance_View LIMIT 0, 1000	1 row(s) returned

26. MINIMUM ACCOUNT BALANCE

The screenshot shows the SQL Developer interface with the following components:

- SQL Editor:** Contains SQL code for creating and querying views.

```
222 • CREATE VIEW Maximum_Balance_View AS
223 SELECT
224     MAX(Balance) AS Maximum_Balance
225 FROM Account;
226 • SELECT * FROM Maximum_Balance_View;
227 • CREATE VIEW Minimum_Balance_View AS
228 SELECT
229     MIN(Balance) AS Minimum_Balance
230 FROM Account;
231 • SELECT * FROM Minimum_Balance_View;
```
- Result Grid:** Displays the result of the last query (line 231).

Minimum_Balance
35000.00
- Output Window:** Shows the execution log with the following entries:

#	Time	Action	Message
40	13:52:02	SELECT * FROM Total_Bank_Balance LIMIT 0, 1000	1 row(s) returned
41	13:53:16	CREATE VIEW Average_Account_Balance AS SELECT ...	0 row(s) affected
42	13:53:16	SELECT * FROM Average_Account_Balance LIMIT 0, 1...	1 row(s) returned
43	13:54:35	CREATE VIEW Maximum_Balance_View AS SELECT ...	0 row(s) affected
44	13:54:35	SELECT * FROM Maximum_Balance_View LIMIT 0, 1000	1 row(s) returned
45	13:55:37	CREATE VIEW Minimum_Balance_View AS SELECT ...	0 row(s) affected
46	13:55:37	SELECT * FROM Minimum_Balance_View LIMIT 0, 1000	1 row(s) returned

27. TOTAL NUMBER OF ACCOUNTS

The screenshot shows the SQL Developer interface with the following components:

- SQL Editor:** Contains SQL code for creating and querying views.

```
227 • CREATE VIEW Minimum_Balance_View AS
228 SELECT
229     MIN(Balance) AS Minimum_Balance
230 FROM Account;
231 • SELECT * FROM Minimum_Balance_View;
232 • CREATE VIEW Account_Count_View AS
233 SELECT
234     COUNT(*) AS Total_Accounts
235 FROM Account;
236 • SELECT * FROM Account_Count_View;
```
- Result Grid:** Displays the result of the last query (line 236).

Total_Accounts
10
- Output Window:** Shows the execution log with the following entries:

#	Time	Action	Message
42	13:53:16	SELECT * FROM Average_Account_Balance LIMIT 0, 1...	1 row(s) returned
43	13:54:35	CREATE VIEW Maximum_Balance_View AS SELECT ...	0 row(s) affected
44	13:54:35	SELECT * FROM Maximum_Balance_View LIMIT 0, 1000	1 row(s) returned
45	13:55:37	CREATE VIEW Minimum_Balance_View AS SELECT ...	0 row(s) affected
46	13:55:37	SELECT * FROM Minimum_Balance_View LIMIT 0, 1000	1 row(s) returned
47	13:56:40	CREATE VIEW Account_Count_View AS SELECT CO...	0 row(s) affected
48	13:56:40	SELECT * FROM Account_Count_View LIMIT 0, 1000	1 row(s) returned

PART E – GROUP BY VIEWS

28. BRANCH-WISE ACCOUNT COUNT

The screenshot shows the SQL Developer interface with the following SQL script in the SQL_VIEWS* window:

```
234 COUNT(*) AS Total_Accounts
235 FROM Account;
236 • SELECT * FROM Account_Count_View;
237 • CREATE VIEW Branch_Account_Count AS
238 SELECT
239     Branch,
240     COUNT(*) AS Number_of_Accounts
241 FROM Account
242 GROUP BY Branch;
243 • SELECT * FROM Branch_Account_Count;
```

The Result Grid displays the following data:

Branch	Number_of_Accounts
Hyderabad	3
Vijayawada	2
Bangalore	1
Chennai	1
Delhi	1
Mumbai	1

The Output window shows the following actions:

#	Time	Action	Message
1	16:57:25	USE BankDB	0 row(s) affected
2	17:02:20	CREATE VIEW Branch_Account_Count AS SELECT Br...	0 row(s) affected
3	17:02:20	SELECT * FROM Branch_Account_Count LIMIT 0, 1000	7 row(s) returned

29. BRANCH-WISE TOTAL BALANCE

The screenshot shows the SQL Developer interface with the following SQL script in the SQL_VIEWS* window:

```
241 FROM Account
242 GROUP BY Branch;
243 • SELECT * FROM Branch_Account_Count;
244 • CREATE VIEW Branch_Total_Balance AS
245 SELECT
246     Branch,
247     SUM(Balance) AS Total_Balance
248 FROM Account
249 GROUP BY Branch;
250 • SELECT * FROM Branch_Total_Balance;
```

The Result Grid displays the following data:

Branch	Total_Balance
Hyderabad	225000.00
Vijayawada	185000.00
Bangalore	120000.00
Chennai	45000.00
Delhi	65000.00
Mumbai	150000.00

The Output window shows the following actions:

#	Time	Action	Message
1	16:57:25	USE BankDB	0 row(s) affected
2	17:02:20	CREATE VIEW Branch_Account_Count AS SELECT Br...	0 row(s) affected
3	17:02:20	SELECT * FROM Branch_Account_Count LIMIT 0, 1000	7 row(s) returned
4	17:03:19	CREATE VIEW Branch_Total_Balance AS SELECT Bra...	0 row(s) affected
5	17:03:19	SELECT * FROM Branch_Total_Balance LIMIT 0, 1000	7 row(s) returned

30. ACCOUNT TYPE COUNT

The screenshot shows the SQL Enterprise Manager interface. The left pane displays the database structure, including the 'bank_db' database and its tables. The right pane shows the SQL script for creating the 'Account_Type_Count' view. The script is as follows:

```
248 FROM Account
249 GROUP BY Branch;
250 SELECT * FROM Branch_Total_Balance;
251 CREATE VIEW Account_Type_Count AS
252 SELECT
253     Account_Type,
254     COUNT(*) AS Number_of_Accounts
255 FROM Account
256 GROUP BY Account_Type;
257 SELECT * FROM Account_Type_Count;
```

The 'Result Grid' shows the output of the query:

Account_Type	Number_of_Accounts
Savings	6
Current	4

The 'Output' pane shows the execution log, including the creation of the view and the selection of data from it.

31. ACCOUNT TYPE TOTAL BALANCE

The screenshot shows the SQL Enterprise Manager interface. The left pane displays the database structure, including the 'bank_db' database and its tables. The right pane shows the SQL script for creating the 'Account_Type_Balance' view. The script is as follows:

```
255 FROM Account
256 GROUP BY Account_Type;
257 SELECT * FROM Account_Type_Count;
258 CREATE VIEW Account_Type_Balance AS
259 SELECT
260     Account_Type,
261     SUM(Balance) AS Total_Balance
262 FROM Account
263 GROUP BY Account_Type;
264 SELECT * FROM Account_Type_Balance;
```

The 'Result Grid' shows the output of the query:

Account_Type	Total_Balance
Savings	355000.00
Current	470000.00

The 'Output' pane shows the execution log, including the creation of the view and the selection of data from it.

PART F – HAVING IN VIEWS

32. BRANCHES WITH MORE THAN ONE ACCOUNT

The screenshot shows the SQL Developer interface with a query window titled 'SQL_VIEWS*'. The query is as follows:

```
263 GROUP BY Account_Type;
264 • SELECT * FROM Account_Type_Balance;
265 • CREATE VIEW Multiple_Account_Branches AS
266 SELECT
267     Branch,
268     COUNT(*) AS Number_of_Accounts
269 FROM Account
270 GROUP BY Branch
271 HAVING COUNT(*) > 1;
272 • SELECT * FROM Multiple_Account_Branches;
```

The 'Result Grid' shows the output of the final SELECT statement:

Branch	Number_of_Accounts
Hyderabad	3
Vijayawada	2

The 'Output' pane shows the 'Action Output' log with the following entries:

#	Time	Action	Message
✓ 5	17:03:19	SELECT * FROM Branch_Total_Balance LIMIT 0, 1000	7 row(s) returned
✓ 6	17:04:00	CREATE VIEW Account_Type_Count AS SELECT Ac...	0 row(s) affected
✓ 7	17:04:00	SELECT * FROM Account_Type_Count LIMIT 0, 1000	2 row(s) returned
✓ 8	17:04:42	CREATE VIEW Account_Type_Balance AS SELECT ...	0 row(s) affected
✓ 9	17:04:42	SELECT * FROM Account_Type_Balance LIMIT 0, 1000	2 row(s) returned
✓ 10	17:05:44	CREATE VIEW Multiple_Account_Branches AS SELECT...	0 row(s) affected
✓ 11	17:05:44	SELECT * FROM Multiple_Account_Branches LIMIT 0, 1...	2 row(s) returned